

Credit - Drishti Ias

UNEP's Limiting Overshoot Report: Managing 1.5°C Breach



For Prelims: United Nations Environment Programme , Paris Agreement , Nationally Determined Contributions , Methane , Global Methane Pledge , Atlantic Meridional Overturning Circulation , Multidimensional Vulnerability Index

For Mains: Global warming and climate mitigation, Climate tipping points and hysteresis, Disaster risk reduction and transformational adaptation

Source: TH

Why in News?

The **United Nations Environment Programme (UNEP)** has released a report titled ***'Limiting Overshoot: Navigating Exceedance of 1.5°C and Pathways Towards Return'*** , which acknowledges that **temporarily breaching the 1.5°C global warming threshold under Paris Agreement , is now virtually inevitable in the near term .**

- The report proposes an **'overshoot, peak, and decline'** framework to limit peak warming and enable a **multi-decadal return towards safer planetary boundaries .**

Summary

- UNEP's Limiting Overshoot report warns that temporarily exceeding the 1.5°C threshold is increasingly inevitable, making it essential to limit peak warming through rapid CO₂ and methane cuts, renewable energy expansion and responsible carbon removal.
- Climate overshoot could trigger irreversible ecosystem changes, sea-level rise, food insecurity and financial vulnerabilities. The report therefore calls for transformational adaptation, stronger Loss and Damage finance, climate justice, ecosystem protection and global cooperation.

What are the Key Highlights of the UNEP "Limiting Overshoot" Report?

- **Imminent Threshold Breach:** Human-induced global warming has reached approximately **1.4°C above pre-industrial levels** , increasing at **~0.25°C per decade** .
 - Operational decadal forecasts indicate a **75% probability** that the five-year mean temperature will exceed **1.5°C within the 2026–2030 period**.
- **Carbon Budget Exhaustion:** To maintain a 50% chance of staying within 1.5°C, the remaining **carbon budget is roughly 130 billion tons of CO₂ from 2026 onwards**.
 - At current emission rates of roughly **40 billion tons of CO₂ per year** , this budget will be completely exhausted in about three years.
- **The Cost of Delay:** Every additional five years of sustained high emissions adds approximately **0.1°C to peak warming** . Cumulatively, emissions since 2015 have already increased the minimum achievable peak warming by **0.2°C** .
- **The Emissions and Implementation Gap:** Under current policies, median projected global warming will reach **2.6°C by 2100** (range: 1.9°C–3.6°C).
 - Even under an optimistic scenario where all **Nationally Determined Contributions (NDCs)** and long-term **Net-Zero targets** are fully executed, central peak warming will still reach **1.8°C** .
- **No Safe Overshoot:** Exceeding 1.5°C is **not benign** ; every fraction of a degree increases the likelihood of crossing irreversible biophysical **tipping points** .
- **Crucial Role of Methane Mitigation:** The UN report highlights **methane mitigation as a critical near-term climate action** , with methane currently contributing around **0.5°C to global warming** .
 - Reducing methane emissions can **slow the rate of warming more rapidly than equivalent CO₂ reductions** due to methane's short **atmospheric lifetime**.
 - While reducing **oil and gas leaks** is relatively feasible, cutting methane from **agriculture and waste** remains challenging. A reduction of **60–70 million tonnes of hard-to-abate methane emissions** could deliver a cooling effect comparable to removing **220 gigatonnes (Gt) of CO₂** .
 - The **Global Methane Pledge** (launched at COP26 by the **European Union** and the US), joined by over **155 countries** , aims to reduce anthropogenic methane emissions by **at least 30% from 2020 levels by 2030** , though major emitters including **India, China, Russia and Iran** have not joined it.
- **Geopolitical Hurdles to Climate Action:**
 - **US Withdrawal:** The United States, the world's largest historical emitter, formally exited the **Paris Agreement and announced its withdrawal from the UNFCCC** , the **IPCC** , and the **Green Climate Fund**.
 - **Global Energy Crisis:** The **2022 conflict in West Asia** has caused the largest oil supply disruption in history, cutting output by over **14 million barrels a day**. Sustained high energy prices could push governments back toward coal and short-term fossil fuel reliance.

What is the 'Overshoot, Peak, and Decline' Pathway?

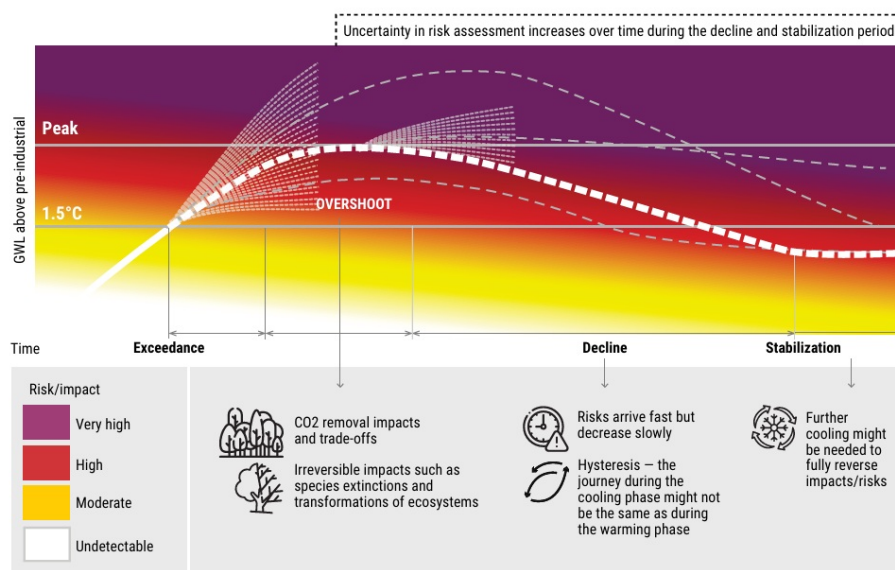
The "overshoot, peak, and decline" pathway shifts the global climate strategy from merely preventing a 1.5°C breach to actively managing it. It involves four sequential stages:

- **Exceedance (Overshoot):** The period in which global warming surpasses the **1.5°C threshold but has not yet reached its peak**.
- **Plateau / Peak:** The period when global warming reaches and plateaus at its maximum level above **1.5°C before starting to fall**.

- **Decline:** The period in which global temperatures begin to steadily decline from their peak level due to global net-negative emissions.
- **Stabilization:** Reached once global temperatures return to or **below the 1.5°C target after the reversal, stabilizing at this safer level.**

Compounding Costs of Breaching 1.5°C (Overshoot)

- **Cryosphere Loss :** By 2100, glaciers outside the **Greenland and Antarctic ice sheets** could lose **25-40% or more of their mass** , contributing around **9 - 15.4 cm to sea-level rise** .
 - Sea-ice recovery can **lag behind temperature recovery** , maintaining lower planetary albedo and increased heat absorption.
- **Sea-Level Rise: Sea levels will continue rising for centuries to millennia** because of deep-ocean warming and ice-sheet inertia.
 - Every **100 years above 1.5°C** could commit the world to approximately **40 cm of additional median sea-level rise by 2300** .
- **Ocean Changes: Ocean acidification and deoxygenation can persist for centuries** because deep-ocean circulation and chemical recovery are extremely slow.
- **Food Security:** Global food production could see declines of up to 14% by 2050 without effective adaptation.
- **Earth-System Tipping Points:**
 - **AMOC:** Sustained warming increases the risk of weakening or collapse of the **Atlantic Meridional Overturning Circulation (AMOC)** , with recovery potentially taking centuries.
 - **Amazon Rainforest: Warming combined with deforestation** could push the **Amazon** from a **carbon sink** towards a **net carbon source** through large-scale forest dieback.
 - **Polar Ice Sheets:** Some parts of the **West Antarctic Ice Sheet** may face irreversible retreat even at relatively low warming levels.



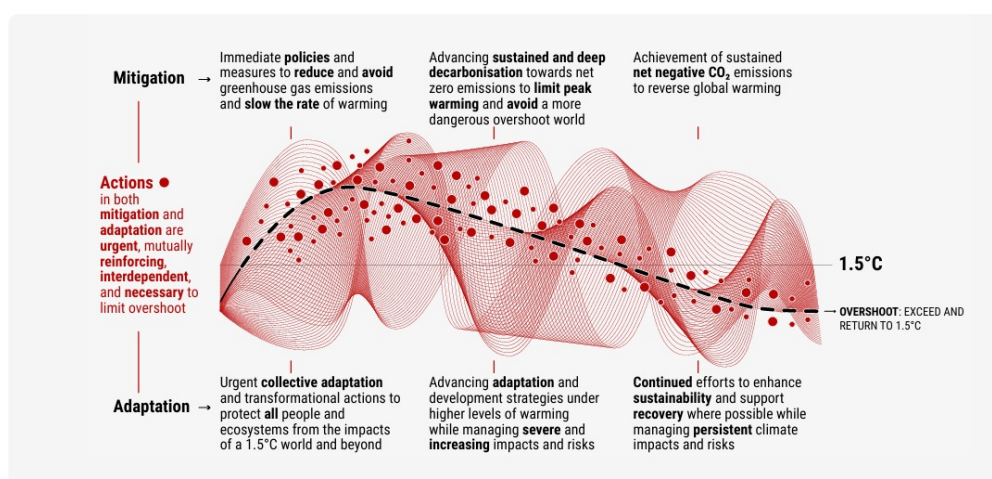
Location of risk colours in this figure are conceptual and not a result of assessment.

What are the UNEP's Key Recommendations for Managing Climate Overshoot?

- **Capitalising the Loss and Damage Architecture: Scale up the Fund for Responding to Loss**

and Damage (FRLD) well before peak warming, while strengthening climate-finance allocation through tools such as the **Multidimensional Vulnerability Index (MVI)** .

- The significant gap between current pledges of around **USD 800 million** and the estimated **USD 395 billion annual needs of developing countries** must be addressed through **non-debt-based financial transfers** .
- Moreover, in low- and middle-income countries, every **USD 1 invested in resilient infrastructure can generate around USD 4 in avoided disaster losses** , highlighting the economic value of **climate-resilient infrastructure** .
- **Accelerate Mitigation:** Pursue **immediate and deep reductions in CO₂ and methane emissions** , as pathways to avoid 1.5°C overshoot are narrowing.
 - Shift long-term national policy from achieving Net-Zero to legally binding **Net-Negative GHG targets** beyond mid-century.
 - Developed economies must hit net-negative emissions first under the principle of **Common But Differentiated Responsibilities and Respective Capabilities (CBDR-RC)** to afford carbon space to developing nations.
- **Scale Renewable Energy:** Increase **renewables to 60-70% of global electricity generation by 2030** , supported by grid expansion, energy storage and demand flexibility.
- **Responsible Scaling of Carbon Dioxide Removal (CDR):** CDR must be deployed in addition to, not instead of, aggressive emission reductions to achieve net-negative emissions.
 - Strong governance frameworks are required to manage the environmental risks, land-use constraints, and scale-up challenges of **both conventional (e.g., afforestation) and novel (e.g., Direct Air Capture) CDR methods**.
- **Preserving SIDS Sovereignty and Entitlements:** Formalize international legal mechanisms ensuring that **Small Island Developing States (SIDS)** retain their legal statehood, sovereignty, and maritime **Exclusive Economic Zones (EEZs)** under **UNCLOS** even if their physical landmass becomes uninhabitable or submerged.
- **Ensure Just Transition:** Integrate employment protection, social security, community participation and equitable distribution of transition costs and benefits.
- **Protect Natural Ecosystems:** Treat **protected areas as climate infrastructure** , strengthening ecosystem protection while preventing **downgrading, downsizing and extractive encroachment** .
- **Strengthen Climate Knowledge:** Invest in **long-term climate observation, monitoring and data systems** , particularly in vulnerable and under-resourced regions.
 - Strengthen the **science-policy interface** by combining **scientific, Indigenous and local knowledge** for continuous climate decision-making.



Conclusion

The **UNEP report** stresses that **breaching 1.5°C reflects past delays but is not a reason for resignation**. The focus must now shift to **simultaneous mitigation and adaptation**, as every fraction of warming avoided, every year of overshoot reduced, and every investment in **climate-resilient infrastructure** can limit planetary risks and irreversible losses.

Drishti Mains Question:

Why does climate overshoot necessitate a transition from incremental to transformational adaptation? Discuss.

Frequently Asked Questions (FAQs)

1. What is the 'Overshoot, Peak and Decline' framework?

It envisages **temporary exceedance of 1.5°C**, followed by a **temperature peak, decline through sustained net-negative emissions, and eventual stabilisation near or below 1.5°C**.

2. Why is methane mitigation important for limiting climate overshoot?

Methane has a **strong short-term warming effect and a shorter atmospheric lifetime than CO₂**, making rapid methane reductions an important lever for slowing near-term warming.

3. What is the role of the Fund for Responding to Loss and Damage (FRLD)?

The **FRLD** is intended to provide financial support to **developing countries particularly vulnerable to climate-related loss and damage**, strengthening climate justice and resilience.

4. Why is Carbon Dioxide Removal (CDR) not a substitute for mitigation?

CDR can help address **residual emissions and achieve net-negative emissions**, but excessive reliance on future removal can create **mitigation deterrence** and delay immediate emission reductions.

5. Why are SIDS particularly vulnerable to climate overshoot?

Sea-level rise, saltwater intrusion, coastal erosion and extreme weather can threaten their habitability and infrastructure, while potentially affecting their **sovereignty and maritime entitlements under UNCLOS**.

[Watch Video on YouTube:

▶ https://www.youtube.com/embed/Y-3X5He_9Fs

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UPSC Civil Services Examination, Previous Year's Question (PYQs)

Prelims :

Q. The term 'Intended Nationally Determined Contributions' is sometimes seen in the news in the context of (2016)

- (a) pledges made by the European countries to rehabilitate refugees from the war-affected Middle East
- (b) plan of action outlined by the countries of the world to combat climate change
- (c) capital contributed by the member countries in the establishment of Asian Infrastructure Investment Bank
- (d) plan of action outlined by the countries of the world regarding Sustainable Development Goals

Answer: (b)

Q. With reference to the Agreement at the UNFCCC Meeting in Paris in 2015, which of the following statements is/are correct? (2016)

1. The Agreement was signed by all the member countries of the UN and it will go into effect in 2017.
2. The Agreement aims to limit the greenhouse gas emissions so that the rise in average global temperature by the end of this century does not exceed 2°C or even 1.5°C above pre-industrial levels.
3. Developed countries acknowledged their historical responsibility in global warming and committed to donate \$1000 billion a year from 2020 to help developing countries to cope with climate change.

Select the correct answer using the code given below.

- (a) 1 and 3 only
- (b) 2 only
- (c) 2 and 3 only
- (d) 1, 2 and 3

Ans: (b)

Mains

Q1. The Intergovernmental Panel on Climate Change (IPCC) has predicted a global sea level rise of about one metre by AD 2100. What would be its impact in India and the other countries in the Indian Ocean region? **(2023)**

Q2. Discuss global warming and mention its effects on the global climate. Explain the control measures to bring down the level of greenhouse gases which cause global warming, in the light of the Kyoto Protocol, 1997. **(2022)**

Q3. Describe the major outcomes of the 26th session of the Conference of the Parties (COP) to the United Nations Framework Convention on Climate Change (UNFCCC). What are the commitments made by India in this conference? **(2021)**

Distinguished Jurist as a Supreme Court Judge



Source: IE

Why in News?

Addressing a convocation at the **National Law University, Delhi** , a **Supreme Court judge** highlighted an **“unused mandate” of the Constitution** : the provision allowing the appointment of a **“distinguished jurist”** to the **Supreme Court** .

- Despite the Constitution being in force for over **76 years** , this provision has **never been used to appoint a legal academic or jurist to the apex court** .

What is the Constitutional Provision Regarding ‘Distinguished Jurist’?

- **Article 124(3) of the Indian Constitution:** It outlines the eligibility criteria for the appointment of a Supreme Court judge. According to the **Article 124(3)**, a person is eligible if they are a citizen of India and fulfill any **one** of the following three conditions:
 - Has been a **High Court judge for at least five years** .
 - Has been an **advocate of a High Court for at least ten years** .
 - Is, **in the opinion of the President** , a **"distinguished jurist"** .
- **For High Court:** For High Court appointments, **Article 217 originally had no such provision** . During the **Emergency** , the **42 nd Constitutional Amendment Act (1976)** introduced a provision **allowing a "distinguished jurist"** to be appointed as a High Court judge.
 - However, this was subsequently repealed by the **44 th Constitutional Amendment Act (1978)** to reverse Emergency-era changes.

Who is a "Distinguished Jurist"?

- **Distinguished Jurist:** The Constitution does not explicitly define the term. However, it is widely understood to mean a person with **exceptional expertise in legal theory, constitutional law, teaching, or jurisprudence** , who has engaged in serious legal scholarship.
 - Crucially, this category allows for the appointment of legal scholars **even if they lack prior judicial or extensive courtroom experience** .
- **Historical Background:**
 - **Constituent Assembly Debates:** The “Distinguished Jurist” clause was introduced during the Constituent Assembly debates in **May 1949** , by **H.V. Kamath** , who argued that the pool of Supreme Court candidates should not be rigidly confined only to **practising advocates and judges**.
 - **Global Precedents:** Judges of the **International Court of Justice (ICJ)** and the US Supreme Court

(such as Harvard professor **Felix Frankfurter** in 1939) were chosen from academia.

- **Dr. B.R. Ambedkar's View:** Dr. Ambedkar supported the idea, though he briefly debated whether the word "**eminent**" would be better suited than "**distinguished**".

Need for Distinguished Jurists in the Supreme Court

- **Diversifying the Bench:** The Supreme Court is not just an adjudicatory body but the nation's "**constitutional conscience-keeper**".
 - Academics can bring diverse perspectives away from standard courtroom conditioning.
- **Overcoming Technicalities:** Jurists often possess a broader macro-view of the law and are less likely to be bogged down by narrow procedural technicalities.
- **Focus on Public Law and Policy:** Supreme Court decisions frequently involve profound constitutional and public-law questions.
 - Renowned scholars can inject deep comparative research, interdisciplinary insights, and theoretical grounding into landmark judgments.

Why has India Never Appointed a Distinguished Jurist?

- **Limited Consideration:** The provision has **not been widely explored in practice by the government and the Collegium**, and the process for identifying suitable candidates remains underdeveloped.
 - Noted legal scholar **Upendra Baxi** once remarked that Presidents have looked through a "telescope" and found no one, effectively making the provision a "dead issue".
- **No Institutional Pathway: Article 217 does not provide a similar route for appointing distinguished jurists to High Courts**, meaning legal academics generally lack a judicial pathway before being considered for the Supreme Court.
- **The Collegium System Hurdle:** Although the Constitution vests the power in the "opinion of the President", modern Supreme Court appointments are initiated by the **Collegium system (headed by the Chief Justice of India)**.
 - An academic would first need to be recommended by the Collegium, which traditionally favors sitting judges and senior advocates.
- **Bar Council Restrictions:** Unlike in the US, **Rule 49 of the Bar Council of India (BCI)** prevents **full-time law teachers from practising law**.
 - This creates a systemic barrier, separating theoretical academic brilliance from practical courtroom engagement.
- **Succession Concerns:** The Supreme Court relies heavily on a seniority-based succession path for appointing the Chief Justice of India. Appointing a younger academic jurist directly to the Supreme Court could disrupt this established hierarchy and affect the succession line.
- **Alternative Utilization of Expertise:** Instead of elevating distinguished jurists to the bench, the Supreme Court frequently utilizes their specialized knowledge by appointing them as ***amicus curiae* (friends of the court)** in highly complex and technical cases.
 - This allows the judiciary to benefit from their academic expertise in a supportive role.
- **The Remuneration Gap:** Legal practice and private consultancies are highly lucrative. Eminent lawyers and top-tier legal scholars would face a massive pay cut if they transitioned to a judicial salary.

- This financial disincentive makes many of the most qualified candidates reluctant to accept a judicial appointment even if it were offered.

Frequently Asked Questions (FAQs)

1. What does Article 124(3) provide regarding a distinguished jurist?

It allows a **citizen of India considered by the President to be a distinguished jurist** to qualify for appointment as a Supreme Court judge.

2. Who is considered a 'distinguished jurist'?

The Constitution does not define the term, but it generally refers to a person with **exceptional expertise in legal scholarship, jurisprudence, constitutional law or legal research**.

3. Has India ever appointed a Supreme Court judge through the distinguished jurist route?

No. Despite the provision existing since the Constitution came into force, the route has remained **unused for over 76 years**.

4. Why is the appointment of distinguished jurists considered significant?

It can **diversify the Bench**, strengthen **constitutional and public-law expertise**, and bring greater academic and interdisciplinary perspectives to judicial decision-making.

5. What is a key challenge in appointing legal academics as Supreme Court judges?

The **Collegium-based appointment process** and **Bar Council restrictions on full-time law teachers practising law** can limit the entry of academics into the higher judiciary.

[Watch Video on YouTube:

▶ <https://www.youtube.com/embed/kudHwR2Apu4>

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UPSC Civil Services Examination, Previous Year Question (PYQ)

Prelims

Q. Consider the following statements: (2019)

1. The 44th Amendment to the Constitution of India introduced an Article placing the election of the Prime Minister beyond judicial review.
2. The Supreme Court of India struck down the 99th Amendment to the Constitution of India as being violative of the independence of judiciary.

Which of the statements given above is/are correct?

- (a) 1 only
- (b) 2 only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

Ans: (b)

Mains

Q. Critically examine the Supreme Court's judgement on the 'National Judicial Appointments Commission Act, 2014' with reference to the appointment of judges of higher judiciary in India. **(2017)**

Hanging Glaciers



Source: IE

Why in News?

Rising temperatures and changing precipitation patterns are destabilising Himalayan glaciers, increasing the **risk of glacier collapse, avalanches and downstream hazards**. A 2026 study identified 219 hanging glaciers in the **Alaknanda basin of the Garhwal Himalayas**, highlighting their growing threat potential.

What are the Key Facts About Hanging Glaciers?

- **About:** Hanging glaciers are **masses of ice located on steep mountain slopes, often detached from the main glacier system** . Due to their unstable position, they can suddenly break off and trigger hazardous ice and rock avalanches.
- **Formation:** **Rapid warming, glacier retreat and climate variability** cause glaciers to lose structural stability. This leads to **redistribution of ice, geometric changes and the development of unstable hanging glacier** formations.
- **Global Context:** Hanging glaciers have been extensively studied in the **Alps** , but basin-scale assessments in the **Himalayas remain limited** due to difficult terrain and inadequate monitoring.

Risks Associated with Hanging Glacier Collapse

- **Avalanche Hazards:** Break-off of hanging glaciers can generate massive ice avalanches. Simulations indicate that avalanche flows in the **Badrinath-Mana sector of Uttarakhand** could exceed **50 metres in height** , potentially damaging settlements and infrastructure.
- **Glacial Lake Outburst Floods (GLOFs):** Glacier collapse can trigger secondary hazards such as **GLOFs** , where sudden release of water from glacial lakes causes destructive downstream flooding.
- **Cascading Hazards:** Collapsing glaciers can **carry rocks, snow and sediments downhill**, generating compound hazards such as debris flows, river blockages and sudden **floods** .
- **Threat to Infrastructure:** Hydropower projects, roads, bridges, settlements and pilgrimage routes located in Himalayan valleys are particularly vulnerable to glacier-related hazards.



Study on Hanging Glaciers in Alaknanda Basin

- **Number of Hanging Glaciers:** An **April 2026 study published in Nature** identified **219 hanging glaciers** in the **Alaknanda basin of the Garhwal Himalayas** .
- **Concentration in Upper Basin:** Nearly one-third of the unstable hanging ice mass is concentrated in the Upper Alaknanda basin, increasing risks for downstream settlements and infrastructure.
- **Increasing Infrastructure Risk:** The Alaknanda basin study projected that avalanche-risk exposure could increase significantly, with **120% more buildings and infrastructure land area at risk by 2030 compared to 2000** .
- **Population Vulnerability:** The population living in these risk-prone areas is projected to increase by around **17% during the same period** , increasing disaster vulnerability.

Frequently Asked Questions (FAQs)

Q1. What are hanging glaciers?

Hanging glaciers are **ice masses located on steep mountain slopes or cliffs** , often disconnected from the main glacier system. Due to their unstable position, they can suddenly detach and trigger **large ice avalanches** .

Q2. Why are hanging glaciers becoming a concern in the Himalayas?

Rising temperatures and changing precipitation patterns are increasing **glacier instability** in the Himalayas. Rapid warming causes changes in glacier movement, ice redistribution, and structural weakening, increasing the possibility of glacier collapse.

Q3. How do hanging glaciers form?

Hanging glaciers develop due to **geometric and dynamic instability** of glaciers. Climate-induced warming alters glacier structure and movement, causing ice masses to become unstable on steep slopes.

Q4. Which Himalayan region has been identified as highly vulnerable to hanging glaciers?

The **Alaknanda basin of the Garhwal Himalayas in Uttarakhand** has been identified as a vulnerable region, with a significant concentration of unstable hanging glaciers.

Q5. What are Glacial Lake Outburst Floods (GLOFs)?

GLOFs are sudden releases of large volumes of water stored in glacial lakes due to the failure of natural ice or moraine dams. They can cause destructive downstream flooding.

[Watch Video on YouTube: [▶ https://www.youtube.com/embed/bdOAWZa65XI](https://www.youtube.com/embed/bdOAWZa65XI)]

UPSC Civil Services Examination, Previous Year Question (PYQ)

Prelims :

Q. Siachen Glacier is situated to the (2020)

- (a) East of Aksai Chin
- (b) East of Leh
- (c) North of Gilgit
- (d) North of Nubra Valley

Ans: (d)

Mains:

Q. With reference to National Disaster Management Authority (NDMA) guidelines, discuss the measures to be adopted to mitigate the impact of the recent incidents of cloudburst in many places of Uttarakhand. **(2016)**

Saptakoteswar Temple



Source: IE

Archaeological remains dating back to the **14th-16th centuries** have been discovered during restoration work at the old site of the **Saptakoteswar Temple** on **Divar Island, Goa** .

- Restoration work along the old pathway between **Koti Tirth and Madhavachi Talli** revealed remains suggesting successive phases of temple construction and reconstruction.
- The discoveries include a **Kirtimukh motif traditionally carved above temple doorways** , a **carved pillar capital** indicating the architectural sophistication of the earlier structure, and **parts of Dev Khostam** used for placing idols within the temple.

Saptakoteswar Temple

- **Shaivite Tradition:** Saptakoteswar Temple is **dedicated to Lord Shiva** and is considered one of the important Shaivite temples of the **Konkan region**.
- **Kadamba Connection:** The temple had strong associations with the **Kadamba dynasty (10th-14th century CE)**, whose rulers were devotees of Lord Shiva. Coins bearing the inscription

“Saptakotishvaralabdha-Varaprasada” reflect the importance of the deity among Kadamba rulers.

- **Vijayanagar Reconstruction:** After destruction during earlier periods, the temple was **reconstructed during the Vijayanagar period**, reflecting continued religious importance.
- **Portuguese Period:** The temple faced **destruction and displacement during Portuguese** rule in Goa, leading to the relocation of the Shiva Linga.
- **Maratha Restoration:** The temple tradition was **revived under Chhatrapati Shivaji Maharaj** , who reconstructed the temple in 1668.
- **Legend Behind the Name:** According to local tradition, seven sages worshipped Lord Shiva at the site, after which Shiva appeared as **Saptakoteswar** (*Sapt* meaning seven and *Koteswar* referring to the Lord of crores).
- **Cultural Significance:** The temple represents Goa’s **layered cultural heritage** , reflecting influences of the **Kadamba, Vijayanagar, Portuguese and Maratha periods** and the continuity of Shaivite traditions in the region.



Read More: [Temple Architecture](#)